

CHAPTER 5 - TUNED CIRCUITS

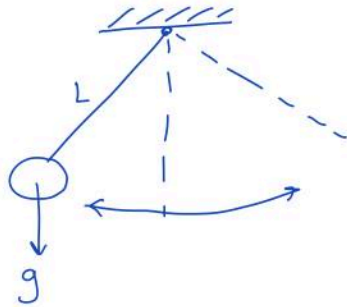
①

* We need frequency selectivity to confine signal to a particular frequency band.

$\Rightarrow$ we use tuned circuits for this

Resonance: tendency of a system to oscillate with greater amplitude at some frequency than at others.

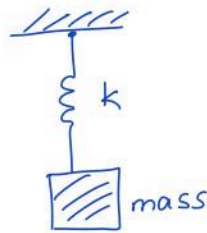
Example: pendulum / swing



$$T \approx 2\pi \sqrt{\frac{L}{g}}$$

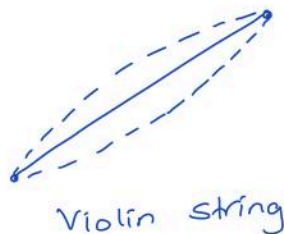
$$\omega = \sqrt{\frac{g}{L}}$$

Example:



$$\omega = \sqrt{\frac{k}{m}}$$

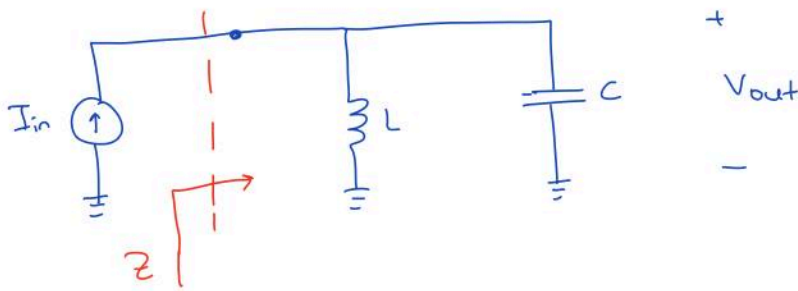
Example:



resonance frequency depends on the length of the string and tension.

Parallel Resonance:

(2)



$$Z = j\omega L \parallel \frac{1}{j\omega C} = \left(\frac{1}{j\omega L} + j\omega C \right)^{-1} = \frac{j\omega L}{1 - \omega^2 LC}$$

When $1 - \omega^2 LC = 0 \Rightarrow Z \rightarrow \infty$

$$\omega L = \frac{1}{\omega C}$$

Capacitor and inductor
cancel each other

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \quad \text{or} \quad f_0 = \frac{1}{2\pi \sqrt{LC}}$$

resonance frequency

* If $i_{in}(t) = I \cdot \cos(\omega_0 t)$

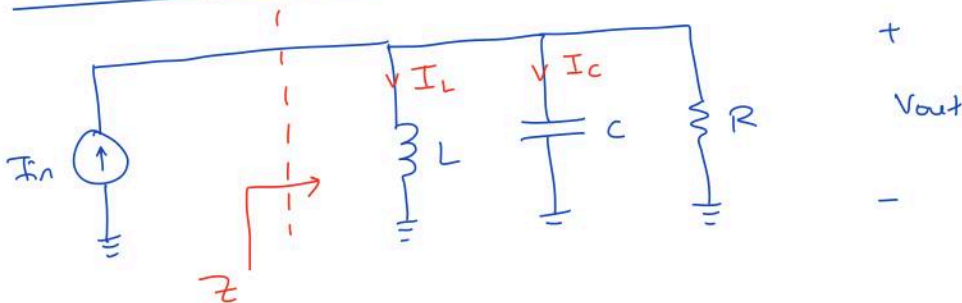
$$V_{out} = Z \cdot I_{in}$$

$$V_{out}(t) = \infty \cdot \cos(\omega_0 t)$$

↑
infinitely large voltage

* This circuit is called a parallel tuned circuit.
This circuit has no resistors $\Rightarrow$ lossless circuit
 $\Rightarrow$ not realistic

more realistic case:



$$Z = j\omega L \parallel \frac{1}{j\omega C} \parallel R = \left(\frac{1}{R} + \frac{1}{j\omega L} + j\omega C \right)^{-1}$$

$$= \frac{j\omega L R}{j\omega L + R - \omega^2 L C R} = \frac{j\omega L}{(1 - \omega^2 LC) + j\omega \frac{L}{R}}$$

$$\textcircled{a} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad , \quad Z = \frac{j\omega_0 L}{j\omega_0 \frac{L}{R}} = R$$

③

* The current source only sees the resistor!

then, $i_{in}(t) = I_p \cdot \cos(\omega_0 t) \Rightarrow I_{in} = I_p$

$$V_{out}(t) = R \cdot I_p \cdot \cos(\omega_0 t) \Rightarrow V_{out} = R \cdot I_p$$

* What is I_L ? What is I_C ? at ω_0

$$I_C = \frac{V_{out}}{\frac{1}{j\omega_0 C}} = j\omega_0 C I_p R$$

$$I_L = \frac{V_{out}}{j\omega_0 L} = \frac{I_p R}{j\omega_0 L} = \frac{I_p R \cdot \omega_0 C (-j)}{j (-j)} = -j\omega_0 C I_p R$$

use $\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow \omega_0 L = \frac{1}{\omega_0 C}$

So, $I_L = -I_C$ @ ω_0

* Stored Energy in Inductor and Capacitor

$$E_C(t) = \frac{1}{2} C V_C^2(t) \quad , \quad \text{where } V_C(t) = V_{out}(t) = R I_p \cdot \cos(\omega_0 t)$$

$$= \frac{1}{2} C R^2 I_p^2 \cos^2(\omega_0 t)$$

$$E_L(t) = \frac{1}{2} L i_L^2(t) \quad , \quad \text{where } I_L = -j\omega_0 C I_p R = \omega_0 C I_p R e^{-j\pi/2}$$

$$i_L(t) = \omega_0 C I_p R \cos(\omega_0 t - \pi/2) = \omega_0 C I_p R \sin(\omega_0 t)$$

So, $E_L(t) = \frac{1}{2} L \omega_0^2 C^2 I_p^2 R^2 \sin^2(\omega_0 t)$

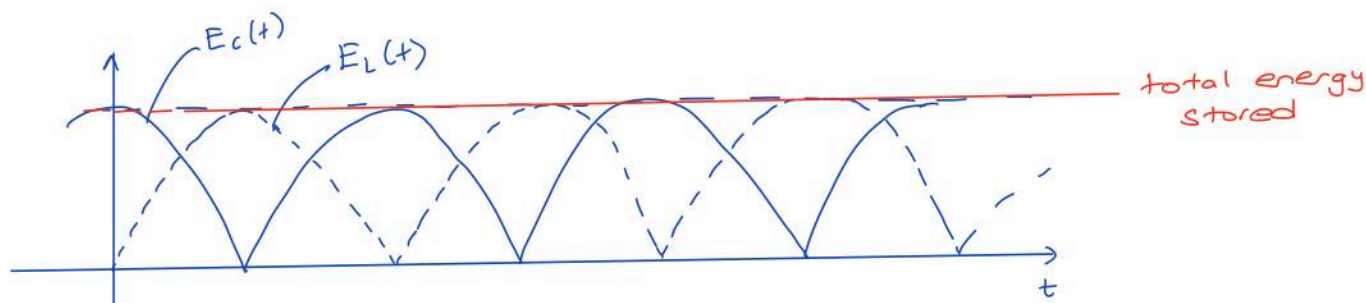
using $\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow \omega_0^2 LC = 1$

$$= \frac{1}{2} C R^2 I_p^2 \sin^2(\omega_0 t)$$

- * When $E_C(t)$ is maximum, $E_L(t) = 0$
 when $E_L(t)$ is maximum, $E_C(t) = 0$
 and,

$$E_C(t) + E_L(t) = \frac{1}{2} C R^2 I_p^2 (\cos^2(\omega_0 t) + \sin^2(\omega_0 t))$$

$$= \frac{1}{2} C R^2 I_p^2, \text{ independent of } t.$$



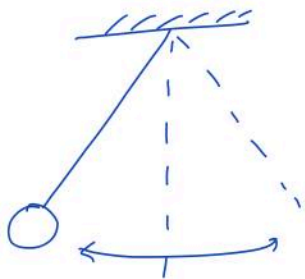
- * Total energy is constant. L & C exchange stored energy.

Quality Factor (Q)

$$Q = \text{quality factor} \propto \frac{\text{stored energy}}{\text{dissipated energy}}$$

if less energy is dissipated $\rightarrow$ higher Q

Example: swing



new swing $\rightarrow$ less friction
 $\rightarrow$ oscillates longer before it stops
 $\rightarrow$ high Q factor

Example: crystal wine glass



hit on wine glass $\rightarrow$ high pitch sound
 if glass is made of high quality crystal,
 sound lasts longer.

$\sim 1 \text{ kHz}$ frequency for sound
 lasts for ~ 5 seconds

$Q \sim 10^3 \times 5 = 5000$
VERY high!!

* Q can be very high in mechanical systems ($Q > 1000$). (5)
 Not so high in electrical systems (10-100), due to resistances / losses.

$$Q = 2\pi \cdot \frac{\text{stored energy}}{\text{energy dissipated in one cycle}}$$

Q for Parallel Resonance Circuit

* Dissipated power: $P_d = \operatorname{Re} \left\{ \frac{V \cdot I^*}{2} \right\} = \operatorname{Re} \left\{ \frac{V_{out} \cdot \left(\frac{V_{out}}{R} \right)^*}{2} \right\}$

where $V_{out} = I_p \cdot R$

So, $P_d = \frac{I_p^2 \cdot R}{2}$

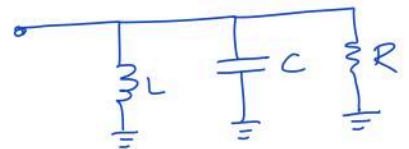
← average power dissipated on the resistor

* Energy dissipated in one cycle = $P_d \cdot T_0$
 $= \frac{I_p^2 \cdot R}{2} \cdot \frac{1}{f_0}$

So,
 $Q = 2\pi \cdot \frac{\frac{1}{2} C R^2 I_p^2}{\frac{I_p^2 R}{2} \cdot \frac{1}{f_0}} = 2\pi f_0 \cdot R C$

$$Q = \omega_0 R C = \frac{R}{\omega_0 L}$$

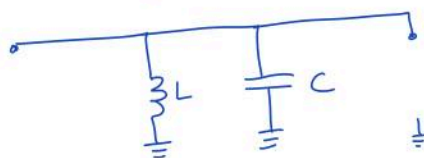
using $\omega_0 C = \frac{1}{\omega_0 L}$



e.g., $\left. \begin{array}{l} \omega_0 = 10,000 \text{ rps} \\ R = 1 \text{ k}\Omega \\ C = 1 \mu\text{F} \end{array} \right\} Q = 10$

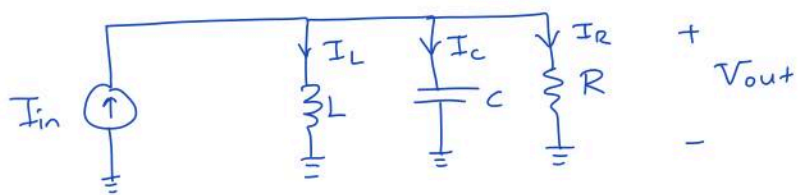
* $Q = \frac{R}{\left| \frac{1}{j\omega_0 C} \right|} = \frac{R}{|j\omega_0 L|}$ (alternative approach)

* Note: Q goes up as R increases $\Rightarrow$ ideally, $R \rightarrow \infty \Rightarrow$ open circuit



← ideal parallel resonance circuit

* Revisit I_L and I_C :

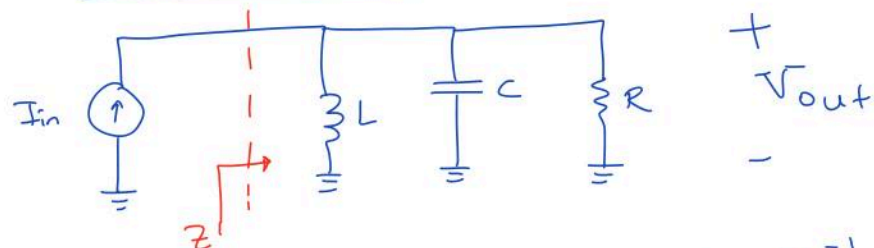


At resonance: $I_R = I_{in} = I_p$
 $V_{out} = R \cdot I_{in}$

$$I_C = \frac{V_{out}}{\frac{1}{j\omega_0 C}} = \underbrace{j\omega_0 C \cdot R}_{Q} \cdot I_{in} = jQ \cdot I_{in}$$

$$I_C = -I_L = jQ I_{in} \quad \left. \vphantom{I_C} \right\} \begin{array}{l} Q \text{ times larger} \\ \text{current than the input!} \end{array}$$

Impedance as a Function of Frequency:



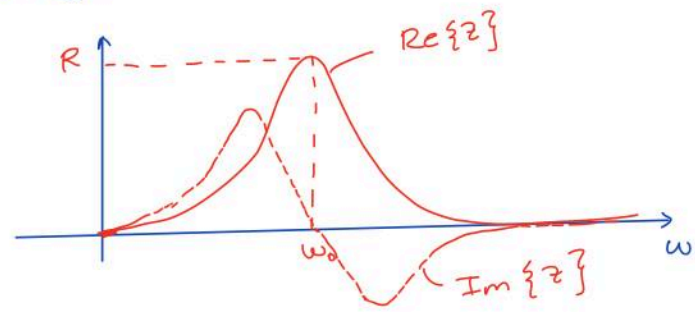
$$Z = \left(\frac{1}{R} + \frac{1}{j\omega L} + j\omega C \right)^{-1} = \frac{j\omega L}{(1 - \omega^2 LC) + j\frac{\omega L}{R}} \quad \begin{matrix} (-j) \\ (-j) \end{matrix}$$

$$= \frac{\omega L}{\frac{\omega L}{R} - j(1 - \omega^2 LC)} \quad \left. \vphantom{Z} \right\} \begin{array}{l} Z \text{ is real-valued} \\ \text{at resonance} \end{array}$$

this is zero at resonance $\omega_0 = \frac{1}{\sqrt{LC}}$

*** Imaginary part of the impedance is zero at resonance.
 $\Rightarrow$ capacitive & inductive components cancel each other.
 $\Rightarrow$ we are left with resistive effects only.

* Voltage and current have the same phase at resonance

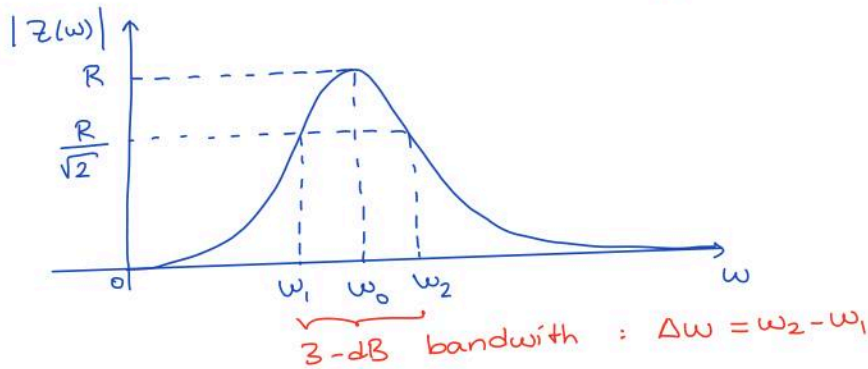


(7)

* Re-write Z using Q :

$$Z = \frac{\frac{wL}{\frac{wL}{R}} - j(1 - w^2LC)}{\frac{R}{wL}} = \frac{R}{1 + j\left(wRC - \frac{R}{wL}\right) \cdot \frac{w_0}{w_0}}$$

$$= \frac{R}{1 + j\left(\underbrace{w_0 RC}_{Q} \cdot \frac{w}{w_0} - \underbrace{\frac{R}{w_0 L}}_{Q} \cdot \frac{w_0}{w}\right)} = \frac{R}{1 + jQ\left(\frac{w}{w_0} - \frac{w_0}{w}\right)}$$



* At some w_1 , $Z(w_1) = \frac{R}{1-j} \Rightarrow |Z(w_1)| = \frac{R}{\sqrt{2}}$

$$Q\left(\frac{w_1}{w_0} - \frac{w_0}{w_1}\right) = -1 \Rightarrow w_1^2 + \frac{w_0}{Q} w_1 - w_0^2 = 0$$

$$w_1 = \frac{-\frac{w_0}{Q} \pm \sqrt{\frac{w_0^2}{Q^2} + 4w_0^2}}{2} \quad \left(\text{we are looking for } w_1 > 0\right)$$

$$w_1 = \frac{-\frac{w_0}{Q} + w_0 \sqrt{\frac{1}{Q^2} + 4}}{2}$$

If $Q > 5$, $w_1 \approx \frac{-\frac{w_0}{Q} + 2w_0}{2} = w_0 - \frac{w_0}{2Q}$

* Similarly, at some w_2 , $Z(w_2) = \frac{R}{1+j} \Rightarrow |Z(w_2)| = \frac{R}{\sqrt{2}}$

$$Q\left(\frac{w_2}{w_0} - \frac{w_0}{w_2}\right) = 1 \Rightarrow w_2^2 - \frac{w_0}{Q} w_2 - w_0^2 = 0$$

$$w_2 = \frac{\frac{w_0}{Q} \pm \sqrt{\frac{w_0^2}{Q^2} + 4w_0^2}}{2} \quad \left(\text{we are looking for } w_2 > 0\right)$$

$$w_2 = \frac{\frac{w_0}{Q} + w_0 \sqrt{\frac{1}{Q^2} + 4}}{2}$$

If $Q > 5$, $\omega_2 \approx \frac{\frac{\omega_0}{Q} + 2\omega_0}{2} = \omega_0 + \frac{\omega_0}{2Q}$

* 3-dB bandwidth:

$$\boxed{\Delta\omega = \omega_2 - \omega_1} = \frac{\frac{\omega_0}{Q} + \cancel{\omega_0 \sqrt{\frac{1}{Q^2} + 4}}}{2} - \frac{-\frac{\omega_0}{Q} + \cancel{\omega_0 \sqrt{\frac{1}{Q^2} + 4}}}{2}$$

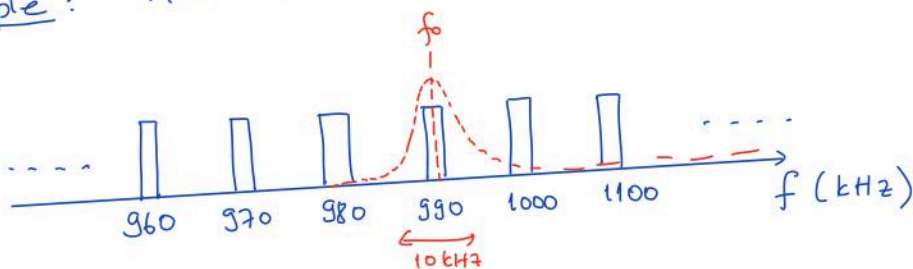
$$\Delta\omega = \frac{\omega_0}{Q} \Rightarrow \boxed{Q = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f}}$$

High $Q \Rightarrow$ narrow bandwidth $\equiv$ good frequency selectivity

* Also, $\boxed{\omega_0 = \sqrt{\omega_1 \cdot \omega_2}}$ (geometric mean)

$$\begin{aligned} \omega_1 \cdot \omega_2 &= \left(\frac{\frac{\omega_0}{Q} + \omega_0 \sqrt{\frac{1}{Q^2} + 4}}{2} \right) \cdot \left(\frac{-\frac{\omega_0}{Q} + \omega_0 \sqrt{\frac{1}{Q^2} + 4}}{2} \right) \\ &= \frac{1}{4} \left[-\cancel{\frac{\omega_0^2}{Q}} + \omega_0^2 \cdot \left(\cancel{\frac{1}{Q^2}} + 4 \right) \right] = \frac{4\omega_0^2}{4} = \omega_0^2 \end{aligned}$$

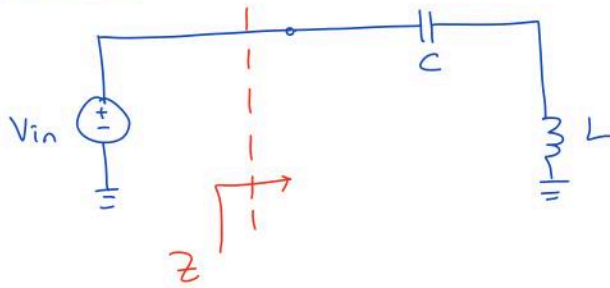
Example: AM radio: channels are 10 kHz apart



$$Q = \frac{f}{\Delta f} \approx \frac{1000 \text{ kHz}}{10 \text{ kHz}} = 100$$

Series Tuned Circuit :

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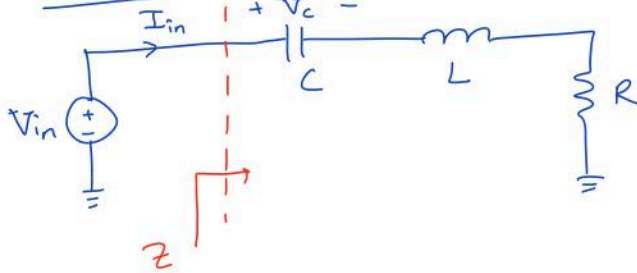
$$Z = \frac{1}{j\omega C} + j\omega L$$

$$= \frac{1 - \omega^2 LC}{j\omega C}$$

* At $(1 - \omega^2 LC) = 0 \Rightarrow Z = 0$ (L & C together act like short circuit)

$\omega_0 = \frac{1}{\sqrt{LC}}$, $f_0 = \frac{1}{2\pi\sqrt{LC}}$: resonance frequency

more realistic case :



$$Z = \frac{1}{j\omega C} + j\omega L + R$$

$$= \frac{(1 - \omega^2 LC) + j\omega RC}{j\omega C} \quad (-j)$$

$$= \frac{\omega RC - j(1 - \omega^2 LC)}{\omega C} \quad (-j)$$

* At resonance, $\text{Im}\{Z\} = 0$:

$1 - \omega^2 LC = 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow Z(\omega_0) = R$

* $V_{in}(t) = V_p \cos(\omega_0 t)$

$i_{in}(t) = \frac{V_p}{R} \cos(\omega_0 t) = I_p \cos(\omega_0 t)$

$V_C = I_{in} \cdot \frac{1}{j\omega_0 C} = I_p \cdot \frac{1}{j\omega_0 C} = \frac{I_p}{\omega_0 C} \cdot e^{-j\pi/2}$

$V_C(t) = \frac{I_p}{\omega_0 C} \cos(\omega_0 t - \frac{\pi}{2}) = \frac{I_p}{\omega_0 C} \sin(\omega_0 t)$

* Energy stored :

$$E_C(t) = \frac{1}{2} C V_C^2(t) = \frac{1}{2} C \frac{I_p^2}{\omega_0^2 C^2} \sin^2(\omega_0 t) \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{2} I_p^2 \cdot L \cdot \sin^2(\omega_0 t)$$

$$E_L(t) = \frac{1}{2} L i_L^2(t) = \frac{1}{2} L i_{in}^2(t) = \frac{1}{2} L I_p^2 \cos^2(\omega_0 t)$$

* Total energy stored:

$$E_{\text{stored}} = E_C(t) + E_L(t) = \frac{1}{2} L I_p^2$$

* Dissipated power: $P_d = \frac{1}{2} I_p^2 R$

$$Q = 2\pi \cdot \frac{\text{stored energy}}{\text{dissipated energy in one cycle}}$$

$$= 2\pi \cdot \frac{\frac{1}{2} L I_p^2}{\frac{1}{2} I_p^2 R \cdot \frac{1}{f_0}} = \frac{2\pi f_0 L}{R}$$

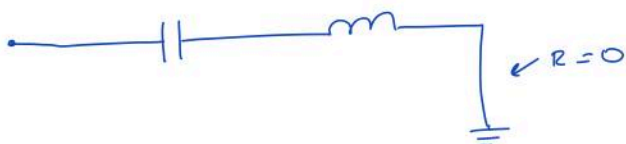
$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 R C}$$

for series tuned circuit

* $Q = \frac{|j\omega_0 L|}{R} = \frac{|\frac{1}{j\omega_0 C}|}{R}$ (alternative approach)

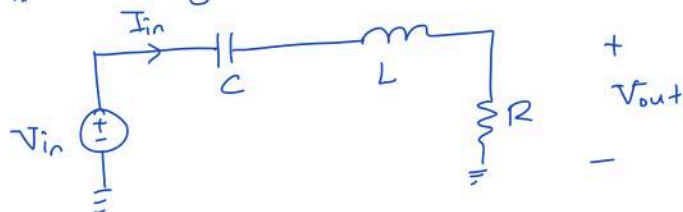
↑ inverse of Q for parallel tuned circuit

* Q goes down if R gets bigger
Ideally, $R=0$



} ideal series tuned circuit.
No power dissipated.

* Voltages on C and L:



at ω_0 , $V_{in} = V_{out}$

$$V_c = I_{in} \cdot \frac{1}{j\omega_0 C} = \frac{V_{in}}{R} \cdot \frac{1}{j\omega_0 C}$$

$$= \frac{V_{in}}{j\omega_0 R C} \quad Q$$

$$V_c = \frac{Q \cdot V_{in}}{j} = -jQ \cdot V_{in}$$

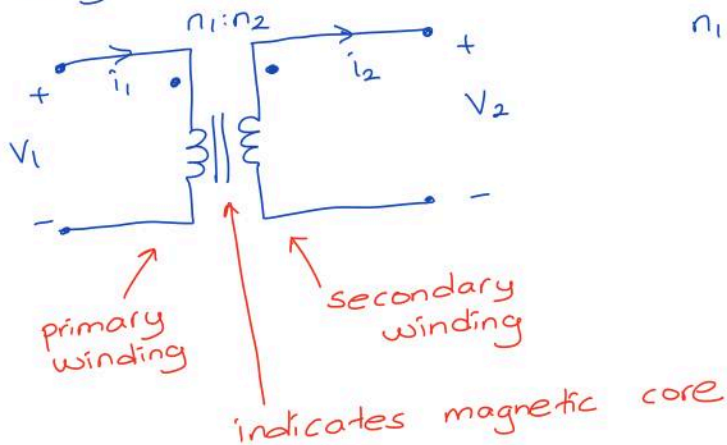
$$V_c = -V_L = -jV_{in} Q \Rightarrow Q \text{ times larger voltage than the input!!}$$

* $Z(\omega) = R \left[1 + jQ \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \right]$ (similar steps as in parallel tuned circuit)

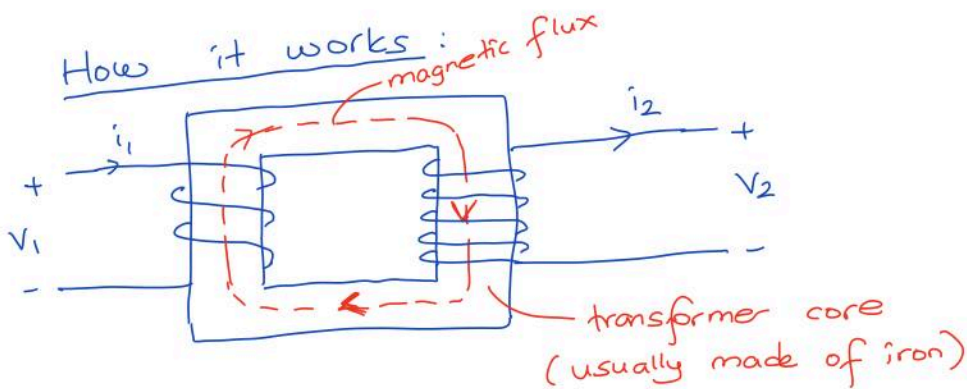
Transformer

- two or more coupled inductors
- they share the same magnetic flux

$n_1 : n_2$ is the turns ratio



How it works:



Shared magnetic flux:

$$V_1 = n_1 \cdot \frac{d\Phi}{dt} \quad \text{flux (from Faraday's Law)}$$

$$V_2 = n_2 \cdot \frac{d\Phi}{dt}$$

$$\text{So, } \boxed{\frac{V_1}{V_2} = \frac{n_1}{n_2}}$$

From conservation of energy, we must have:

$$P_1 = P_2$$

$$V_1 \cdot i_1 = V_2 \cdot i_2$$

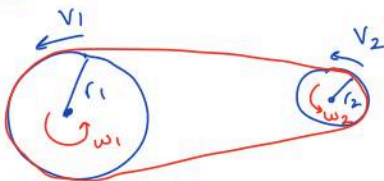
$$\text{So, } \frac{i_1}{i_2} = \frac{V_2}{V_1} = \frac{n_2}{n_1}$$

$\Rightarrow$

$$\boxed{\frac{\hat{i}_1}{\hat{i}_2} = \frac{n_2}{n_1}}$$

* Transforms the voltage and current amplitudes, based on the turns ratio.

Similar: bicycle gear



$$v_1 = v_2$$

$$\omega_1 r_1 = \omega_2 r_2$$

$$\frac{\omega_1}{\omega_2} = \frac{r_2}{r_1}$$

(angular velocity is analogous to current)

$$P_{in} = P_{out}$$

$$T_1 \times \omega_1 = T_2 \times \omega_2$$

$\Rightarrow$

$$\frac{T_1}{T_2} = \frac{\omega_2}{\omega_1}$$

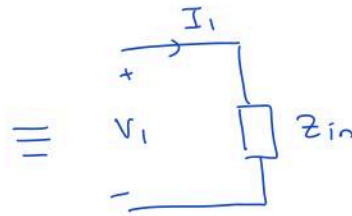
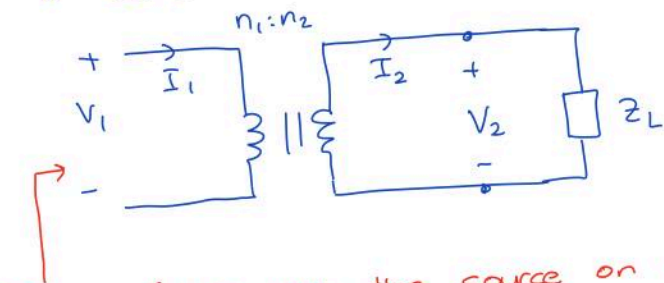
(torque is analogous to voltage)

Impedance Transformation using Ideal Transformers

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Transformers: most common impedance transformation devices.

* When connected to a load:



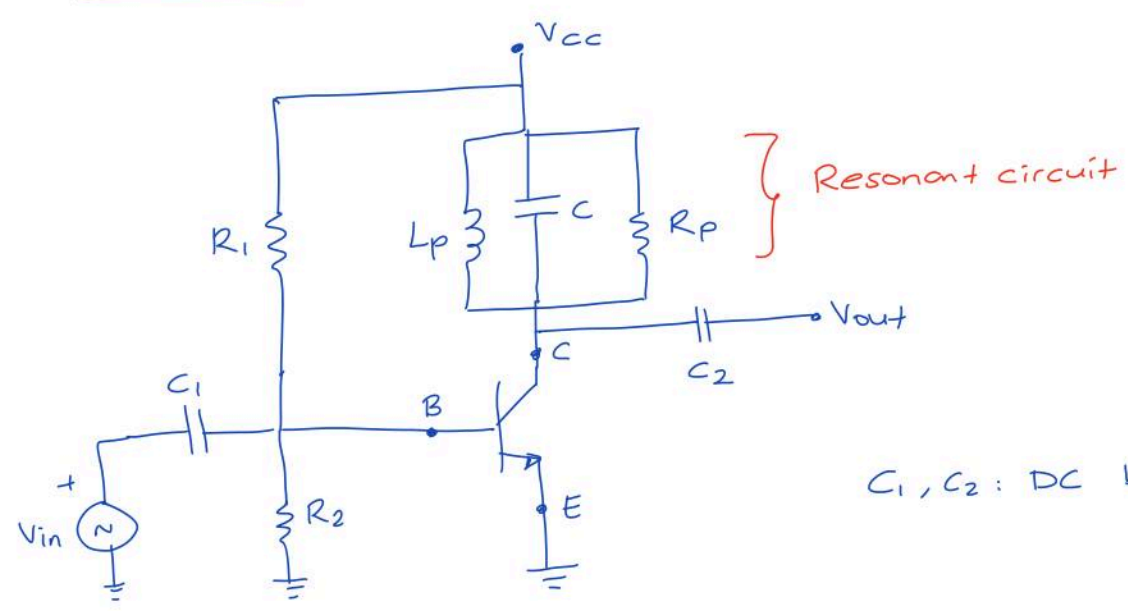
Z_{in} (the way the source on the left sees the load)

$$Z_{in} = \frac{V_1}{I_1}, \quad Z_L = \frac{V_2}{I_2} = \frac{V_1 \left(\frac{n_2}{n_1} \right)}{I_1 \left(\frac{n_1}{n_2} \right)} = \underbrace{\frac{V_1}{I_1}}_{Z_{in}} \cdot \left(\frac{n_2}{n_1} \right)^2$$

$$\boxed{Z_{in} = \frac{V_1}{I_1} = \left(\frac{n_1}{n_2} \right)^2 Z_L}$$

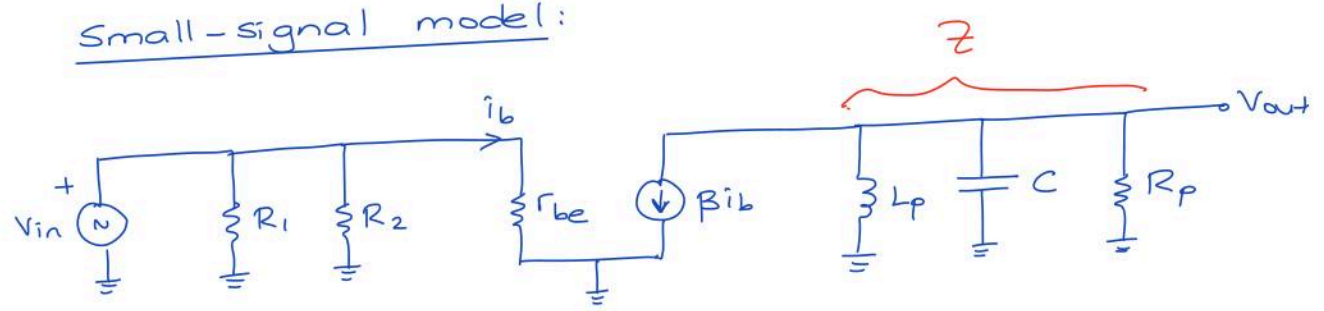
* By changing $\frac{n_1}{n_2}$, we can scale Z_L arbitrarily.

Tuned Amplifiers



C_1, C_2 : DC blocking capacitors

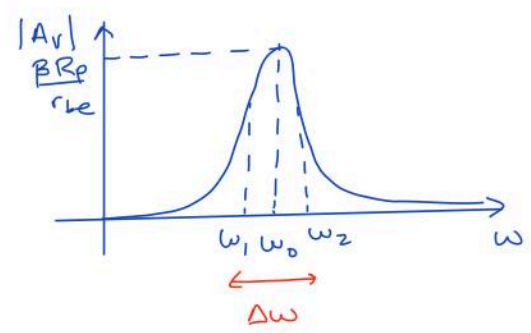
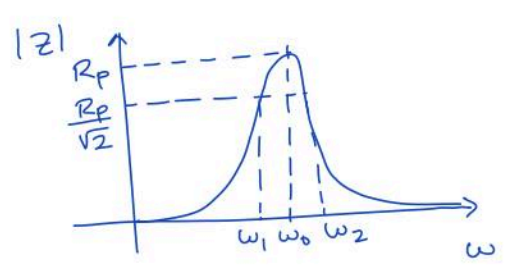
Small-signal model:



$$r_{be} = \frac{25.9 \text{ mV}}{I_B (\text{mA})}$$

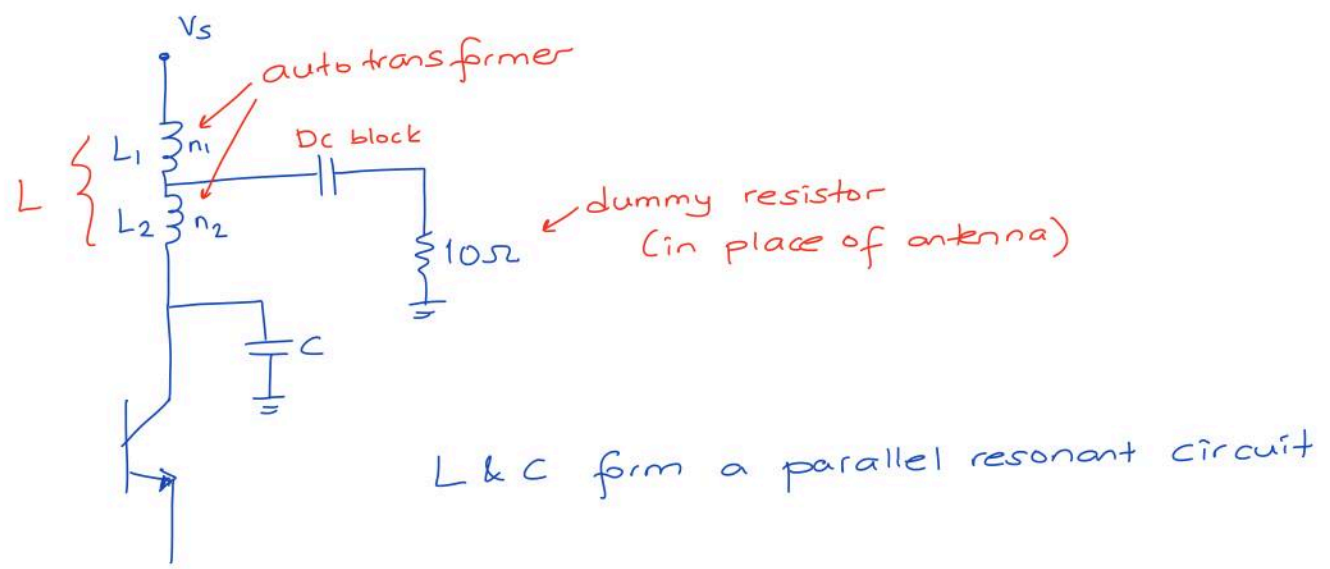
$$i_b = \frac{V_{in}}{r_{be}}, \quad v_{out} = -\beta i_b \cdot Z = -\beta \cdot \frac{V_{in}}{r_{be}} \cdot Z$$

* So, voltage gain is: $|A_v| = \frac{|V_{out}|}{|V_{in}|} = \beta \frac{|Z|}{r_{be}}$

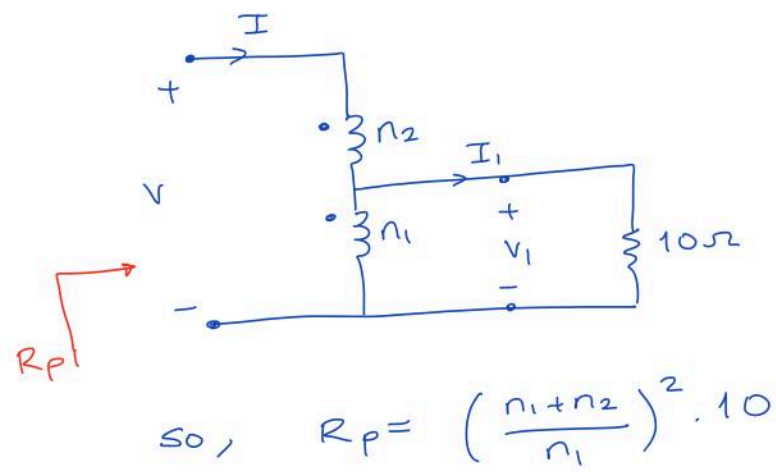


$$Q = \frac{\omega_0}{\Delta \omega}$$

In TRC-II :



Auto transformer : transformer with primary & secondary connected together

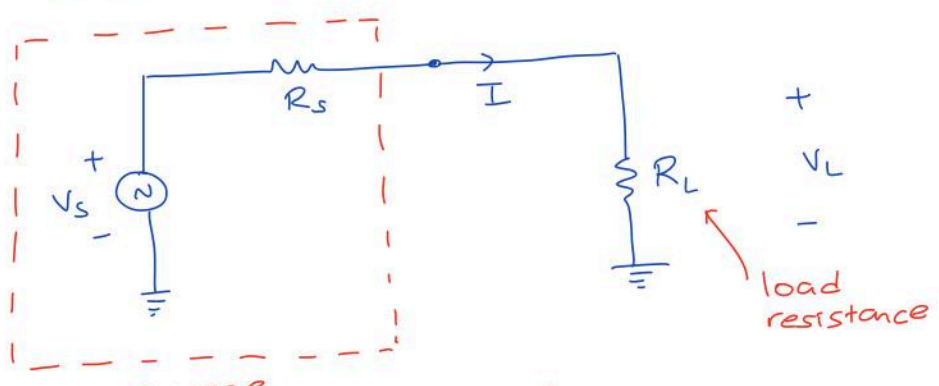


$$\frac{V}{V_1} = \frac{n_1 + n_2}{n_1}$$

$$\frac{I}{I_1} = \frac{n_1}{n_1 + n_2}$$

} transforms small antenna resistance to large R_p (~100-200Ω) to get large voltage gain

Maximum Power Transfer



What load resistance R_L will result in maximum power transfer at the load?

source
(also Thevenin eq. model of a linear circuit)

$$I = \frac{V_s}{R_s + R_L}$$

$$P_L = \frac{|I|^2}{2} \cdot R_L = \frac{1}{2} \frac{|V_s|^2}{(R_s + R_L)^2} \cdot R_L$$

* To find R_L that maximizes power:

$$\frac{\partial P_L}{\partial R_L} = 0 \Rightarrow \frac{\partial P_L}{\partial R_L} = \frac{|V_s|^2}{2} \frac{(R_s + R_L)^2 - R_L \cdot 2(R_s + R_L)}{(R_s + R_L)^4}$$

$$\frac{|V_s|^2}{2} \frac{R_s^2 - R_L^2}{(R_s + R_L)^4} = 0$$

$R_L = R_s$

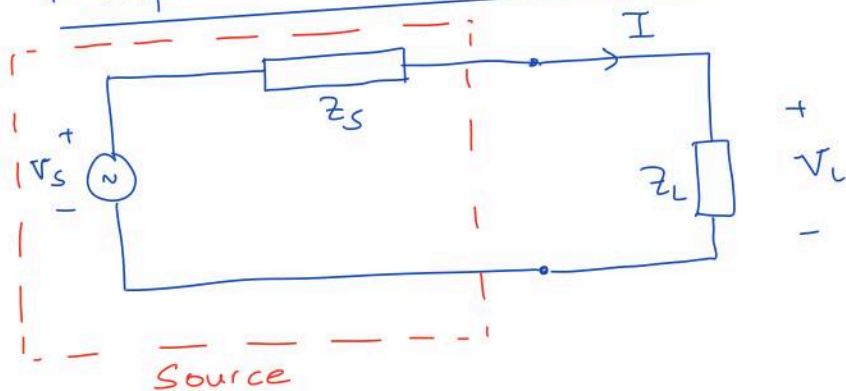
for maximum power transfer

Then,

$$P_L = \frac{1}{2} \frac{|V_s|^2}{(R_s + R_s)^2} \cdot R_s = \frac{|V_s|^2}{8 R_s} = P_A$$

power available at the source

* If the source impedance is complex:



(Thevenin equivalent model of a linear circuit)

$$I = \frac{V_s}{Z_s + Z_L}$$

$$P_L = \operatorname{Re} \left\{ \frac{1}{2} V_L I^* \right\}$$

$$P_L = \operatorname{Re} \left\{ \frac{1}{2} I \cdot Z_L \cdot I^* \right\}$$

$$P_L = \operatorname{Re} \left\{ \frac{1}{2} |I|^2 Z_L \right\} = \operatorname{Re} \left\{ \frac{1}{2} \frac{|V_s|^2}{|Z_s + Z_L|^2} \cdot (R_L + jX_L) \right\}$$

where $Z_L = R_L + jX_L$
 $Z_s = R_s + jX_s$

$$P_L = \frac{1}{2} \frac{|V_s|^2}{|Z_s + Z_L|^2} \cdot R_L = \frac{1}{2} \frac{|V_s|^2}{(R_s + R_L)^2 + (X_s + X_L)^2} \cdot R_L$$

$R_s \geq 0, R_L \geq 0,$
 X_s and X_L could be negative or positive

* To maximize P_L , we make $(X_s + X_L) = 0$

$$\boxed{X_L = -X_s}$$

Then, $P_L = \frac{1}{2} |V_s|^2 \cdot \frac{R_L}{(R_s + R_L)^2} \Rightarrow$ same problem as when we had resistors only

maximum at $\boxed{R_L = R_s}$

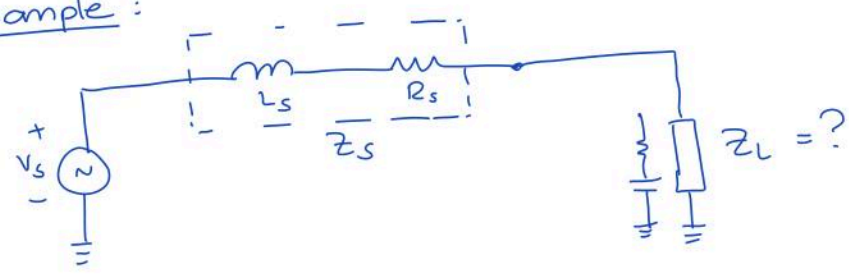
* So,

$$Z_L = R_s - jX_s = Z_s^*$$

$$\boxed{Z_L = Z_s^*} \text{ for maximum power transfer}$$

Then, $P_{L, \max} = \frac{|V_s|^2}{8R_s} = P_A$ ← available power from source

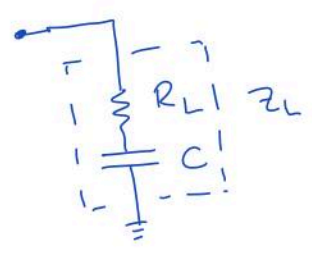
Example :



$$Z_s = R_s + j\omega L_s$$

$$Z_L = Z_s^* = R_s - j\omega L_s$$

we can use a capacitor to get negative reactance



$$Z_L = R_L + \frac{1}{j\omega C} = R_s - j\omega L_s$$

$$R_L - \frac{j}{\omega C} = R_s - j\omega L_s$$

$$\boxed{R_L = R_s}$$

$$\frac{1}{\omega C} = \omega L_s \Rightarrow \boxed{C = \frac{1}{\omega^2 L_s}}$$

Use a capacitor to cancel out the inductor at a specific frequency ω .

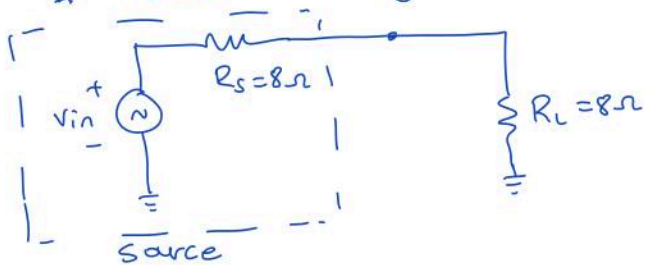
Example: 1000 W audio amplifier to be used in a concert.

$R_S = 8\Omega$. However, we have a speaker

that has a load of $R_L = 3\Omega$.

How much power do we lose by using this speaker instead of $R_L = 8\Omega$ speaker?

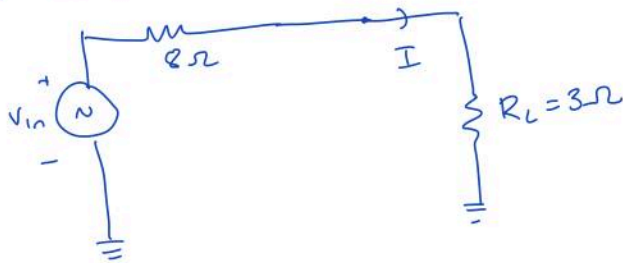
* Power ratings are specified assuming $R_L = R_S$:



$$P_{L,max} = P_A = \frac{|V_{in}|^2}{8R_S}$$

$$V_{in} = \sqrt{8R_S \cdot P_{L,max}} = 253 \text{ V} \quad \left(\begin{array}{l} \text{peak voltage} \\ \text{of the} \\ \text{sinusoid} \end{array} \right)$$

* What we have is:

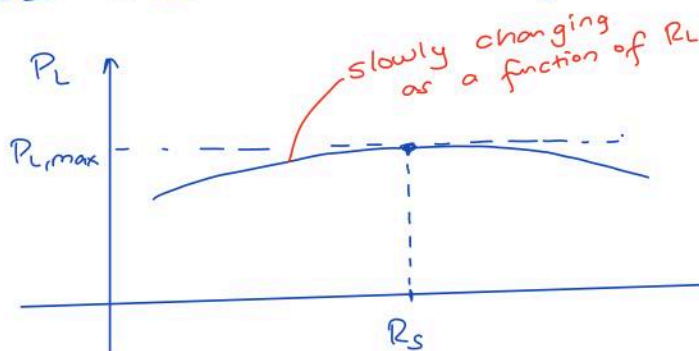


$$P_L = \frac{1}{2} |I|^2 \cdot R_L$$

$$= \frac{1}{2} \frac{|V_{in}|^2}{(R_S + R_L)^2} \cdot R_L$$

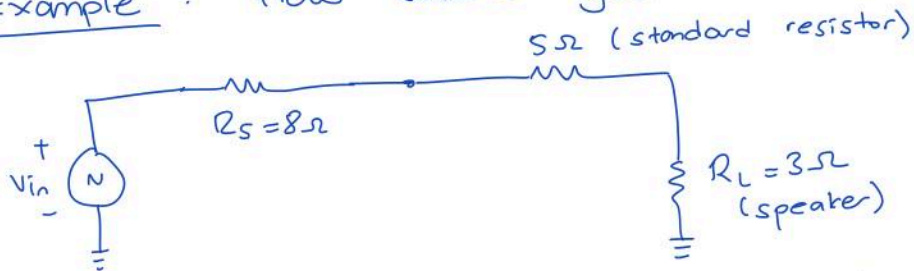
$$= \frac{1}{2} \frac{253^2}{11^2} \cdot 3 = 793 \text{ W}$$

* We lose 207 W of the available 1000 W power.



$$P_L = \frac{1}{2} \frac{|V_{in}|^2}{(R_S + R_L)^2} \cdot R_L$$

Example: How would you solve this problem?

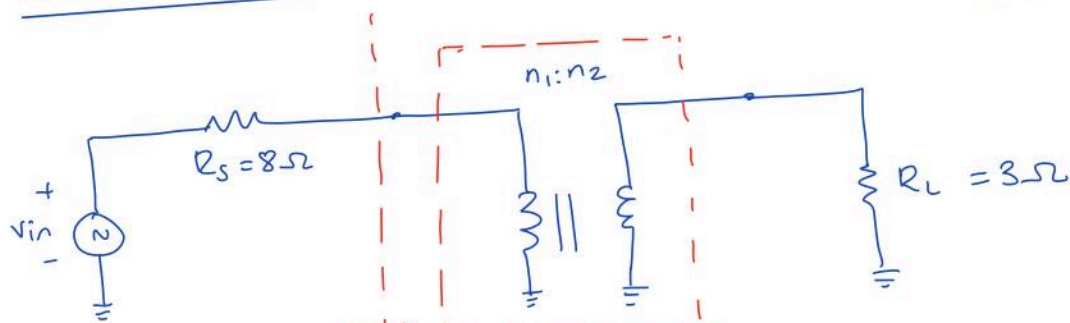


Is this a solution?

makes the mismatch worse: $(8+5)\Omega$ vs. 3Ω

$$P_L = \frac{1}{2} \frac{|V_{in}|^2}{(13+3)^2} \cdot 3 = 375 \text{ W}$$

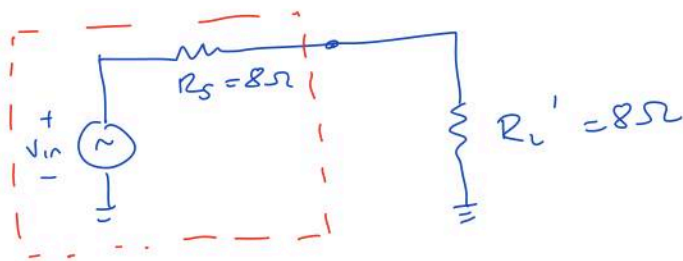
One solution: make the source "see" the 3Ω load as an 8Ω load.



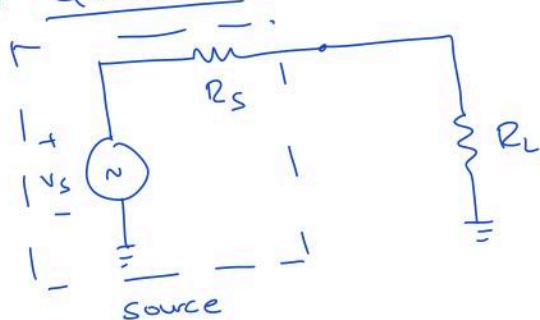
$$R_{eq} = \left(\frac{n_1}{n_2}\right)^2 \cdot R_L = 8\Omega \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{8}{3}} \approx 1.63$$

and we would deliver all 1000 W to the load.

What the source sees:



* Question:



Assume R_L is fixed.

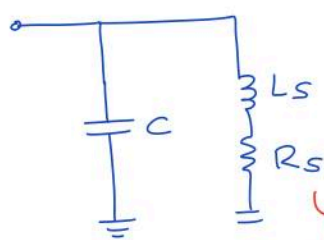
If we are designing the source, how should we choose R_s ?

$$P_L = \frac{1}{2} \frac{|V_s|^2}{(R_s + R_L)^2} R_L$$

try to make this as low as possible.
Ideally, $R_s = 0$.

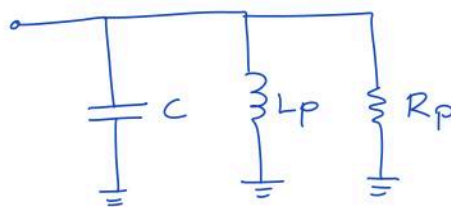
Realistic LC Parallel Circuit

(20)



around
resonance freq
 ω_0
 $\approx$

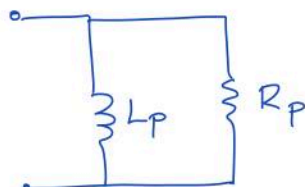
internal
resistance
of inductor



$$Q = \frac{\omega_0 L_s}{R_s}$$



$\approx$



$$Q = \frac{R_p}{\omega_0 L_p}$$

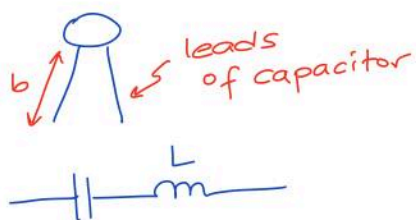
$$Q = \frac{\omega_0 L_s}{R_s} = \frac{R_p}{\omega_0 L_p}$$

$$R_p = R_s (1 + Q^2)$$

$$L_p = L_s \left(1 + \frac{1}{Q^2}\right)$$

$$\omega_0 = \frac{1}{\sqrt{L_p C}}$$

Real Capacitors

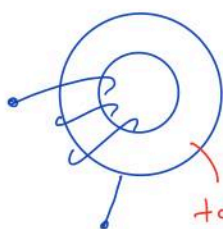


$$L \text{ (nH)} = 2b \left[\ln \left(\frac{2b}{r} \right) - 0.75 \right]$$

length of wire in cm
radius of wire in cm

* cut the leads of the capacitor short to reduce inductive effects

Real Inductors:



toroidal
powdered iron core

$$L = A_L N^2$$

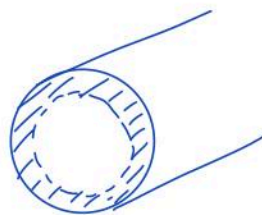
of turns

inductance constant
depends on geometry and the core

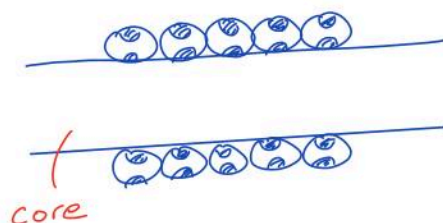
* Loss in inductors $\rightarrow$ core loss
 $\rightarrow$ copper loss

* $R = \rho \frac{l}{A}$: at DC
 (ρ - resistivity, l - length, A - area)

* At higher frequencies:



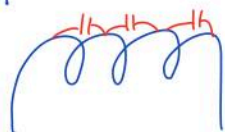
current flows close to the skin of the wire
 $\Rightarrow$ skin effect
 $A_{eff} \downarrow$, $R \uparrow$



proximity effect
 $A_{eff} \downarrow$, $R \uparrow$

RF inductors are wound as single layer inductors

* Self resonance in inductors:



(parasitic capacitance)

} has a self resonance freq.

Use the inductor below the self resonance